Pizza Sales Analysis using SQL

Data-Driven Business Insights

DATA ANALYSIS

SQL

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Project Overview

Objective

Analyze pizza sales data end-to-end using SQL to surface revenue trends, demand patterns, and customer preferences.

Dataset

Four relational tables: **orders**, **order_details**, **pizzas**, and **pizza_types** — linked by `order_id` and `pizza_id`.

Goal

Turn raw transactional data into clear business insights that drive smarter menu and marketing decisions.

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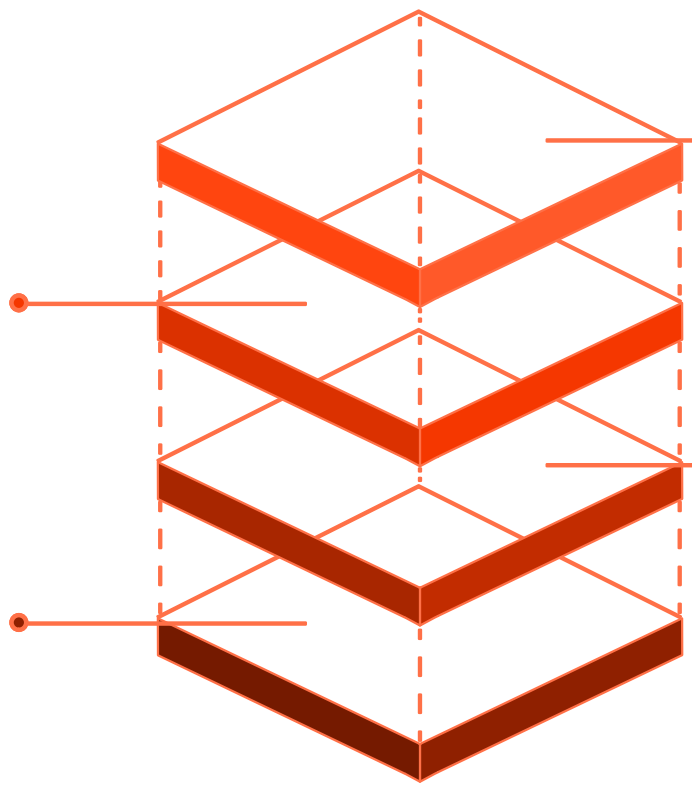
Database Schema

Four tables form the core relational model, connected through `order_id` and `pizza_id` foreign keys.



Pizza Types

Describes pizza_type_id, name, category and ingredients

Pizzas

Defines pizza_id, size, price

Order Details

Links order_id to pizza_id and quantity

Orders

Tracks order_id, date, time

This normalized schema enables multi-table `JOIN` queries that link customer orders directly to pizza attributes and pricing — the foundation for every analysis in this project.

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Total Orders

SQL Query

```
SELECT COUNT(order_id) AS total_orders
FROM orders;
```

Insight

Over **21,000 orders** processed signals strong, sustained customer demand — a healthy baseline for deeper revenue and product analysis.

Total Orders

21,350

Orders Placed

Full fiscal year

High order volume confirms a loyal, active customer base driving consistent revenue.

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Total Revenue

SQL Query

```sql
SELECT ROUND(
  SUM(order_details.quantity
   * pizzas.price), 2)
  AS total_revenue
FROM order_details
JOIN pizzas
  ON pizzas.pizza_id
   = order_details.pizza_id;
```

Insight

Nearly **$818K in revenue** generated from consistent order flow — averaging roughly **$38 per order**. Strong unit economics across the full menu.

Total Revenue

$817.8K

Annual Revenue

Across all pizza orders

Consistent order flow drives predictable, scalable revenue.

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Highest Priced Pizza

SQL Query

```sql
SELECT pizza_types.name,
       pizzas.price
FROM pizza_types
JOIN pizzas
  ON pizza_types.pizza_type_id
   = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Insight

At **$35.95**, The Greek Pizza commands a premium price point — **nearly 2× the menu average**. Upselling strategies around specialty items can meaningfully lift profit margins.

🏆 Top Premium Item

$35.95

The Greek Pizza

Highest priced on the menu

Premium items represent untapped margin opportunity.

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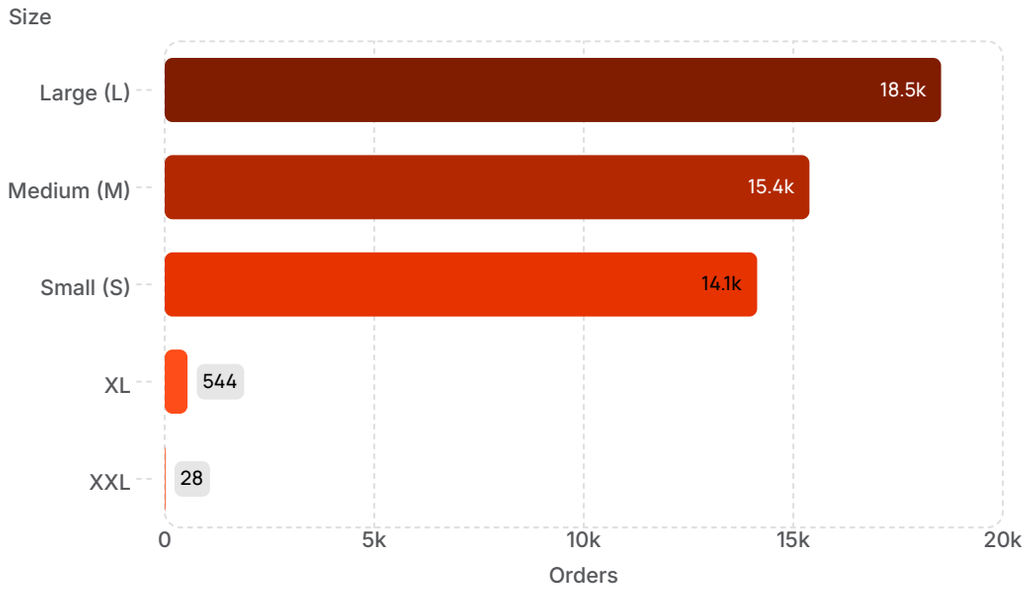
Most Ordered Pizza Size



SQL Query

```
SELECT pizzas.size,
  COUNT(order_details
   .order_details_id)
   AS order_count
FROM order_details
JOIN pizzas
  ON order_details.pizza_id
   = pizzas.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Large (L) is the dominant size at 18,526 orders — **38.1% of all orders**. Bundle promotions and combo deals around Large pizzas will capture the highest demand segment.

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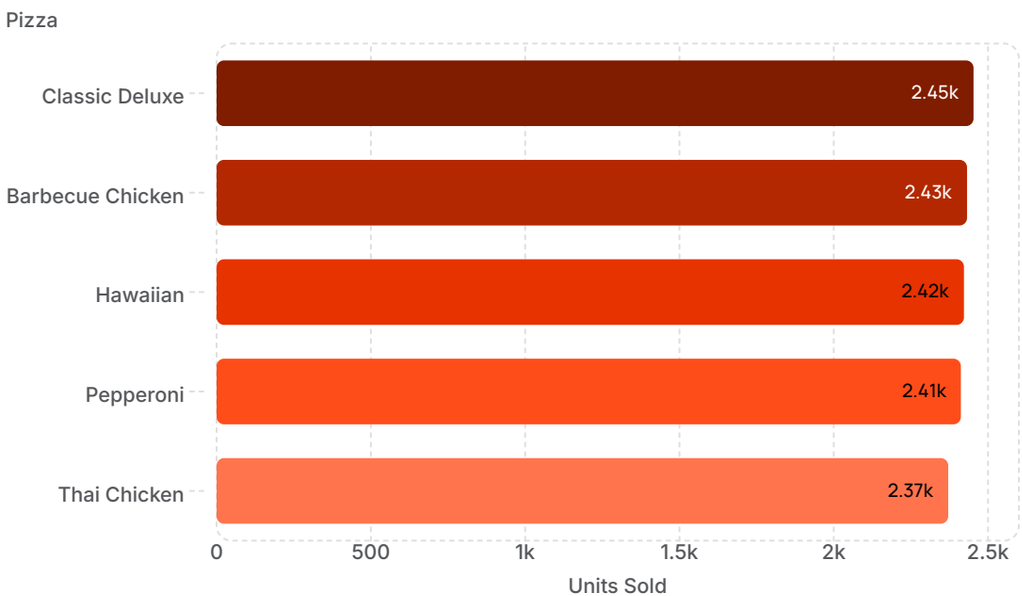
Top 5 Best-Selling Pizzas



SQL Query

```
SELECT pizza_types.name,
  SUM(order_details.quantity)
  AS quantity
FROM pizza_types
JOIN pizzas ON
  pizza_types.pizza_type_id
  = pizzas.pizza_type_id
JOIN order_details ON
  order_details.pizza_id
  = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

These 5 SKUs account for a disproportionate share of volume — a classic **80/20 pattern**. Prioritizing inventory and marketing around these items maximizes ROI.

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Key Insights & Recommendations

What the Data Shows

Large pizzas dominate — 38% of all orders; size preference is clear and consistent.

Top 5 pizzas drive the majority of unit volume — revenue is highly concentrated.

Premium items underutilized — The Greek Pizza's $35.95 price point shows margin headroom.

Recommended Actions

→ **Promote top 5 pizzas** with featured placements and limited-time offers to sustain sales momentum.

→ **Launch Large-size combos** — bundle deals tailored to the dominant size preference.

→ **Upsell premium pizzas** via staff training and digital menu callouts to lift average order value.

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Conclusion

This project demonstrates how **structured SQL queries** transform raw transactional data into actionable business intelligence.

SQL Proficiency

Multi-table JOINs, aggregations, GROUP BY, ORDER BY, and LIMIT applied across real-world datasets.

Business Acumen

Every query was framed around a concrete business question — revenue, demand, pricing, and product mix.

Analytical Thinking

Findings translated directly into strategic recommendations — not just numbers, but decisions.

📬 Open to Data Analyst opportunities — **connect via LinkedIn or portfolio**

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